

Q5. Agreement Between Exploratory Findings and Model-Based Results

A key objective of this assignment is to assess whether the conclusions drawn from the exploratory analysis in Assignment 1 are consistent with those obtained from the fitted longitudinal models. The exploratory phase suggested substantial heterogeneity in BPRS trajectories over time, as well as a non-trivial variance structure that appeared to evolve across follow-up. In this section, these findings are compared with the results from the selected linear mixed-effects model (LMM), with particular emphasis on the mean structure, between-subject heterogeneity, and variance behavior.

Agreement in Mean Structure and Heterogeneity

The exploratory analysis in Assignment 1 revealed a clear time trend in BPRS scores and pronounced between-patient variability in both baseline levels and longitudinal evolution. These features are directly reflected in the final LMM, which includes both a patient-specific random intercept and a patient-specific random slope for time. Model comparison showed a substantial improvement in fit when the random slope was added, as evidenced by a large reduction in -2 restricted log-likelihood, AIC, and BIC (Table 5). This confirms that the heterogeneity observed during exploration represents a genuine structural feature of the data rather than random noise.

Thus, with respect to the mean trajectory and between-subject heterogeneity, the exploratory and model-based results are in strong agreement.

Empirical Variance Function Versus Model-Implied Variance

A more subtle diagnostic concerns the agreement between the empirical variance observed during exploration and the variance implied by the fitted LMM. The empirical variance of BPRS, computed separately at each time point, shows a decreasing pattern over time (Figure Y, blue curve). In contrast, the marginal variance implied by the fitted LMM increases with time (Figure Y, red curve), even under the assumption of homoscedastic residual errors.

As shown in Figure 2, the empirical variance of BPRS decreases over time, whereas the model-implied marginal variance increases due to the contribution of patient-specific random slopes. Allowing for time-specific residual variances does not materially improve agreement, indicating that the discrepancy is driven by selection effects and trajectory heterogeneity rather than residual heteroscedasticity.

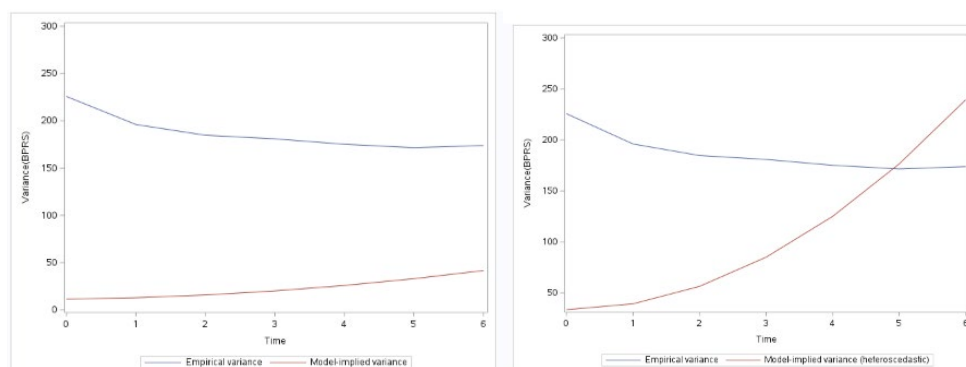


Figure 2. Empirical versus model-implied marginal variance of BPRS over time. Left: Empirical variance (blue) and variance implied by the selected linear mixed-effects model with random intercept and slope assuming homoscedastic residuals (red). Right: Empirical variance (blue) and variance implied by a model allowing time-specific residual variances (red).

This apparent discrepancy arises from the fundamentally different nature of the two quantities. The empirical variance is a descriptive measure computed directly from the observed data at each visit and reflects a combination of biological heterogeneity, measurement error, and changes in the composition of the observed sample over time. In the present study, monotone dropout likely leads to selective retention of patients with more stable or less extreme BPRS values, providing a plausible explanation for the observed decline in empirical variance.

By contrast, the model-implied variance from the LMM represents the marginal variance induced by the assumed random-effects structure. Even with constant residual variance, the inclusion of a random slope for time induces a time-dependent marginal variance of the form

$$\text{Var}(Y_{ij}) = \sigma_{b0}^2 + 2t_{ij}\sigma_{b0b1} + t_{ij}^2\sigma_{b1}^2 + \sigma^2,$$

where the increasing contribution of the random slope leads to growing dispersion of latent individual trajectories over time. Consequently, the fitted variance reflects divergence of subject-specific trajectories rather than the observed spread at each time point.

Assessment of Heteroscedastic Residuals

To examine whether the discrepancy between empirical and fitted variance could be attributed to heteroscedastic residual errors, the LMM was extended to allow time-specific residual variances using a variance components structure. The estimated residual variances were identical across time points (Table Z), indicating that the data do not support heteroscedastic residual variability. As a result, the extended model effectively reduces to the homoscedastic residual model, and the fitted variance curve remains essentially unchanged (Figure 2).

Table 9. *Estimated time-specific residual variances from the heteroscedastic linear mixed-effects model for BPRS.*

Obs	CovParm	Group	Estimate
37	Residual	time 0	20.8725
38	Residual	time 1	20.8725
39	Residual	time 2	20.8725
40	Residual	time 3	20.8725
41	Residual	time 4	20.8725
42	Residual	time 5	20.8725
43	Residual	time 6	20.8725

This finding suggests that the observed discrepancy is not due to misspecification of the residual variance, but instead reflects the conceptual difference between unconditional empirical variability and the parametric variance structure imposed by the LMM.

Is the Discrepancy a Problem?

The discrepancy between empirical and model-implied variance functions is not inherently problematic for the objectives of the analysis. The primary inferential interest lies in the fixed effects describing the mean trajectory and in the characterization of between-subject heterogeneity, both of which are strongly supported by the data and robust to model extensions. The use of Kenward–Roger degrees-of-freedom adjustments further mitigates potential consequences of variance misspecification for fixed-effect inference.

Such a discrepancy would become problematic only if it led to materially different conclusions regarding fixed effects or if accurate modeling of the variance structure itself were a primary objective. In the present context, neither condition applies. Instead, the variance comparison provides diagnostically useful insight into the role of dropout and selection effects, reinforcing conclusions drawn from the missingness analysis in Assignment 1.

Conclusion

In summary, the exploratory findings from Assignment 1 are largely consistent with the results of the fitted linear mixed-effects model. The strong improvement in fit obtained by including random slopes confirms the heterogeneity in individual BPRS trajectories observed during exploration. Although the empirical variance function differs from the model-implied variance, this discrepancy is well understood, informative, and does not undermine inference on the mean structure. Overall, the fitted LMM provides an appropriate and robust representation of the key longitudinal features of the data.

